Solution

Sinusoids

Copyright © Barry Van Veen 2016

AllSignalProcessing.com



1. Frequency is _____ proportional to fundamental period.

   a) directly

   b) inversely

2. Continuous-time sinusoids with distinct frequencies are _____.

   a) always distinct

   b) not necessarily distinct

3. Discrete-time sinusoids with distinct frequencies are _____.

   a) always distinct

   b) not necessarily distinct

4. Discrete-time frequency is measured in units of _____.

   a) Hz

   b) rads/sec

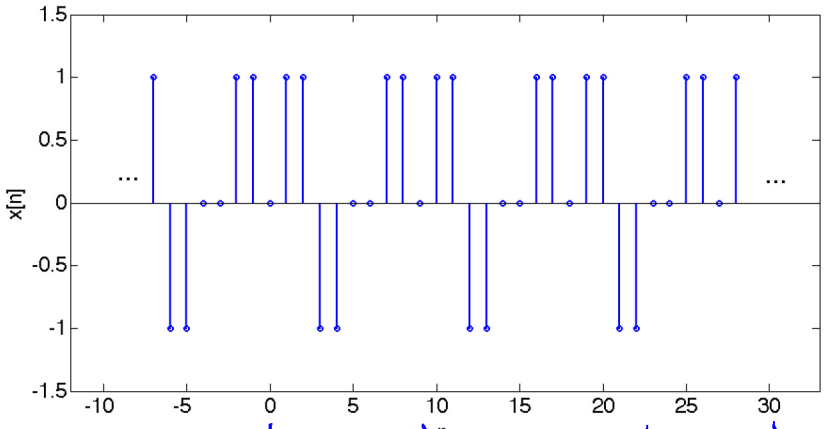
   c) rads

5. A complex sinusoid $x[n] = Ae^{j\pi n/8}$ has real part given by a _____.

   a) cosine

   b) sine

6. Find the fundamental frequency of the signal depicted below.



N=9 Fundamental period N=9

fundamental frequency = $\frac{2\pi}{9}$

7. Identify which of the following discrete-time sinusoids are identical

a) $x[n] = 5\cos(\frac{\pi}{4}n)$

b) $x[n] = 5\cos(\frac{3\pi}{8}n)$

c) $x[n] = 5\cos(\frac{19\pi}{8}n)$

d) $x[n] = 5\cos(\frac{19\pi}{4}n)$

e) $x[n] = 5\cos(\frac{9\pi}{8}n)$

f) $x[n] = 5\cos(\frac{21\pi}{8}n)$

$\cos(\omega_1 n) = \cos(\omega_2 n)$ if $\omega_2 = \omega_1 \pm \ell\, 2\pi$ for ℓ integer

$\frac{19\pi}{8} = \frac{3\pi}{8} + \underbrace{\frac{16\pi}{8}}_{2\pi}$

Verify your answer using DTSinusoid.m.

8. A continuous-time sinusoid $x(t) = 4\cos(\Omega_o t + \pi/8)$ is sampled to produce a discrete-time sinusoid $x[n] = x(nT)$. Find the frequency ω_o of the discrete-time sinusoid for the following choices of Ω_o and T.

a) $\Omega_o = 4\pi$ rads/sec, $T = 0.1$ secs

b) $\Omega_o = 4\pi$ rads/sec, $T = 0.2$ secs

c) $\Omega_o = \pi$ rads/sec, $T = 0.1$ secs

d) $\Omega_o = 100\pi$ rads/sec, $T = 10^{-3}$ secs

$x[n] = 4\cos(\Omega_o nT + \frac{\pi}{8})$

let $\omega_0 = \Omega_0 T$

then $x[n] = 4\cos(\omega_0 n + \frac{\pi}{8})$

a) $\omega_0 = 0.4\pi$ rads

b) $\omega_0 = 0.8\pi$ rads

c) $\omega_0 = 0.1\pi$ rads

d) $\omega_0 = 0.1\pi$ rads

9. The fundamental frequency of a trumpet note $x(t)$ is f_c Hz.

a) Find the fundamental frequency in units of rads/sec if $f_c = 880$ Hz.

b) You sample $x(t)$ at an interval $T = 10^{-4}$ sec to obtain $x[n]$. What is the fundamental frequency of $x[n]$ if $f_c = 880$ Hz.

c) You sample $x(t)$ at an interval $T = 10^{-4}$ sec to obtain $x[n]$. What is the fundamental frequency of $x[n]$ if $f_c = 1760$ Hz.

a) $\Omega_c = 2\pi f_c = 1760\pi$ rads/sec

b) $\omega_c = \Omega_c T = 0.1760\pi$ rads

c) $\omega_c = 2\pi f_c T = 0.3520\pi$ rads

10. Write $x(t) = 4\cos(120\pi t + \pi/3)$ as a sum of two complex sinusoids.

use $\cos(\theta) = \frac{e^{j\theta} + e^{-j\theta}}{2}$

$x(t) = 2e^{j(120\pi t + \pi/3)} + 2e^{-j(120\pi t + \pi/3)}$

11. Write $x[n] = je^{j\pi n/8} - je^{-j\pi n/8}$ as a real-valued sinusoid.

$\sin(\theta) = \frac{e^{j\theta} - e^{-j\theta}}{2j}$ $\quad x[n] = -\frac{e^{j\pi n/8} + e^{-j\pi n/8}}{j}$

$x[n] = -2\sin(\pi n/8)$

12. Find the fundamental period for each of the following signals.

a) $x(t) = \cos(\pi t)$

b) $y(t) = \sin(2\pi t/3)$

c) $z(t) = x(t) + y(t)$

use $\cos(\theta) = \cos(\theta + 2\pi)$

$\sin(\theta) = \sin(\theta + 2\pi)$

Hint: if a signal $x(t)$ has fundamental period T_x, then it also is periodic with period mT_x where $m = 1, 2, 3, \ldots$.

a) $x(t) = \cos(\pi t) = \cos(\pi t + 2\pi) = \cos(\pi(t+2))$

$= x(t+2)$ $\quad T_x = 2$ sec

b) $y(t) = \sin(2\pi t/3) = \sin(2\pi t/3 + 2\pi) = \sin(\frac{2\pi}{3}(t+3))$

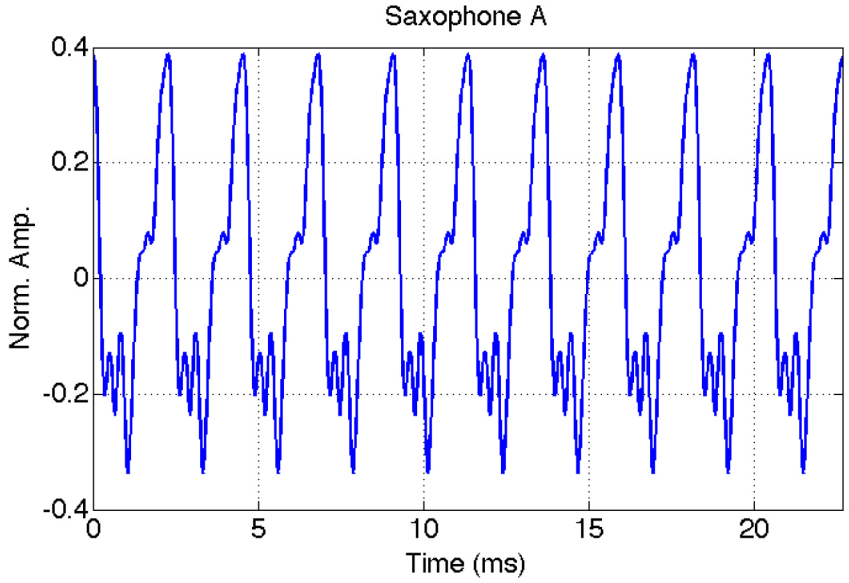
$= y(t+3)$ $\quad T_y = 3$ sec

c) $x(t)$ has periods $2, 4, 6, 8, 10, \ldots$

$y(t)$ has periods $3, 6, 9, 12, \ldots$

minimum common period $T_z = 6$ sec

13. The sound file SaxA.wav contains a brief recording of an alto saxophone playing the musical note A. Listen to the sound. A 22.65 ms section of this note is depicted below.



Use the graph to estimate the fundamental frequency of this note.

10 periods in 22.65 mS $\Rightarrow$

Fundamental period is 2.265 mS

Fund. frequency $f_0 = \frac{1}{2.265 \times 10^{-3}} = 441$ Hz

14. The m-file SaxAapprox.m will display the approximation of this note as a sum of eight harmonically related cosine signals. Signals are harmonically related if they are integer multiples of a common fundamental frequency. The kth harmonic of fundamental f_o has frequency kf_o. Listen to the approximations using 1, 2, 3, 7, and 8 sinusoidal components and compare them to the original. Use the data cursor to find the approximate amplitude of the following cosines in the approximation.

a) $k = 2$ 0.067

b) $k = 5$ 0.034

